

Question

Tungsten nitrides have attracted significant attention from scientists due to their excellent chemical, mechanical, and thermal properties. However, incorporating nitrogen into the tungsten lattice is thermodynamically unfavorable under atmospheric pressure, making the preparation of tungsten nitrides challenging and requiring relatively harsh conditions. Scientists have synthesized a series of nitrides by reacting sodium tungstate with boron nitride under high temperature and high pressure, among which M is one. M crystallizes in the hexagonal system with unit cell parameters $a = 289 \text{ pm}$, $c = 1529 \text{ pm}$, and a tungsten-nitrogen bond length of $d = 216 \text{ pm}$. If A, B, C represent the close-packed layers of nitrogen atoms, and a, b, c represent the close-packed layers of tungsten atoms, with $\square$ denoting a vacant layer, then the stacking sequence of this crystal is $\dots AbAbA\square BaBaB\square AbAbA\square BaBaB\square \dots$.

Based on the given information, select the correct option from the following choices.

- A.** According to the above data, the density of M is $6.15 \text{ g} \cdot \text{cm}^{-3}$
- B.** In the unit cell of M, nitrogen atoms are packed to form octahedral voids, tetrahedral voids, and trigonal prismatic voids. The numbers of these voids in one unit cell are 2, 4, and 8, respectively. Additionally, the coordination numbers of both nitrogen and tungsten atoms in M are 6.
- C.** In the chemical equation for the reaction of sodium tungstate with boron nitride to produce the product, the ratio of the sum of the coefficients on the reactant side to the sum of the coefficients on the product side is 12:11.
- D.** The interlayer spacing between the two nitrogen atom layers above and below the vacancy layer $\square$ in M crystal is 187 pm .

Answer

Correct Answer: **C**

Detailed Explanation

A. The structure contained within one unit cell of the M crystal is "AbAbA□BaBaB□", where uppercase letters represent N and lowercase letters represent W. Thus, a unit cell contains 6 N atoms and 4 W atoms, with the chemical formula W_2N_3 . The density

$$\rho = \frac{zM}{N_A V} = \frac{6 \times 14.01 + 4 \times 183.8}{6.02 \times 10^{23} \times \frac{\sqrt{3}}{2} \times 289^2 \times 1529 \times 10^{-30}} \text{ g} \cdot \text{cm}^{-3} = 12.3 \text{ g} \cdot \text{cm}^{-3}$$

Therefore, option A is incorrect.

CHECKPOINT

1 PTS

$$\rho = 12.3 \text{ g} \cdot \text{cm}^{-3}$$

B. Identical N atom layers can form trigonal prismatic voids A(b/c)A or B(a/c)B. Each such arrangement generates two such trigonal prismatic voids in the unit cell, totaling $4 \times 2 = 8$ in one unit cell. Tetrahedral voids A(a/b)B and octahedral voids A(c)B exist between layers stacked with different letters, numbering $2 \times 2 = 4$ and $2 \times 1 = 2$, respectively. All W atoms are in trigonal prismatic coordination environments with a coordination number of 6. The N atoms have two coordination environments: one surrounded by W atoms in a trigonal prismatic arrangement (coordination number 6), and the other adjacent to one layer of W and one empty layer (coordination number 3). The ratio of N atoms with these two coordination numbers is 1:2. Thus, option B is incorrect.

CHECKPOINT

1 PTS

Nitrogen atom stacking forms octahedral voids, tetrahedral voids, and trigonal prismatic voids, numbering 2, 4, and 8 per unit cell, respectively

CHECKPOINT

1 PTS

All W atoms are in trigonal prismatic coordination environments with a coordination number of 6

CHECKPOINT

1 PTS

N atoms have two coordination environments with coordination numbers of 6 and 3

C. Under high temperature and pressure, sodium tungstate reacts with boron nitride to produce W_2N_3 as well as stable products corresponding to Na, O, and B, such as sodium metaborate $NaBO_2$, or stable products corresponding to B and O, such as B_2O_3 . Excess nitrogen escapes as N_2 . Since the only reactant containing Na and O is sodium tungstate Na_2WO_4 , and the Na-to-O ratio is 1:2 (matching $NaBO_2$), B_2O_3 should not appear as a separate product. The balanced equation is:



The sum of reactant coefficients is 12, and the sum of product coefficients is 11. Thus, option C is correct.

CHECKPOINT

2 PTS



D. The height of the tungsten-coordinated trigonal prism:

$$h = 2 \times [d^2 - (a/\sqrt{3})^2]^{1/2} = 2 \times [(216 \text{ pm})^2 - (289 \text{ pm}/\sqrt{3})^2]^{1/2} = 274 \text{ pm}$$

CHECKPOINT

1 PTS

$$h = 274 \text{ pm}$$

Thus, the interlayer spacing of nitrogen atoms:

$$d_{\text{N-N}} = (c - 4h)/2 = (1529 \text{ pm} - 4 \times 274 \text{ pm})/2 = 216 \text{ pm}$$

CHECKPOINT

1 PTS

$$d_{\text{N-N}} = 216 \text{ pm}$$

Therefore, option D is incorrect.